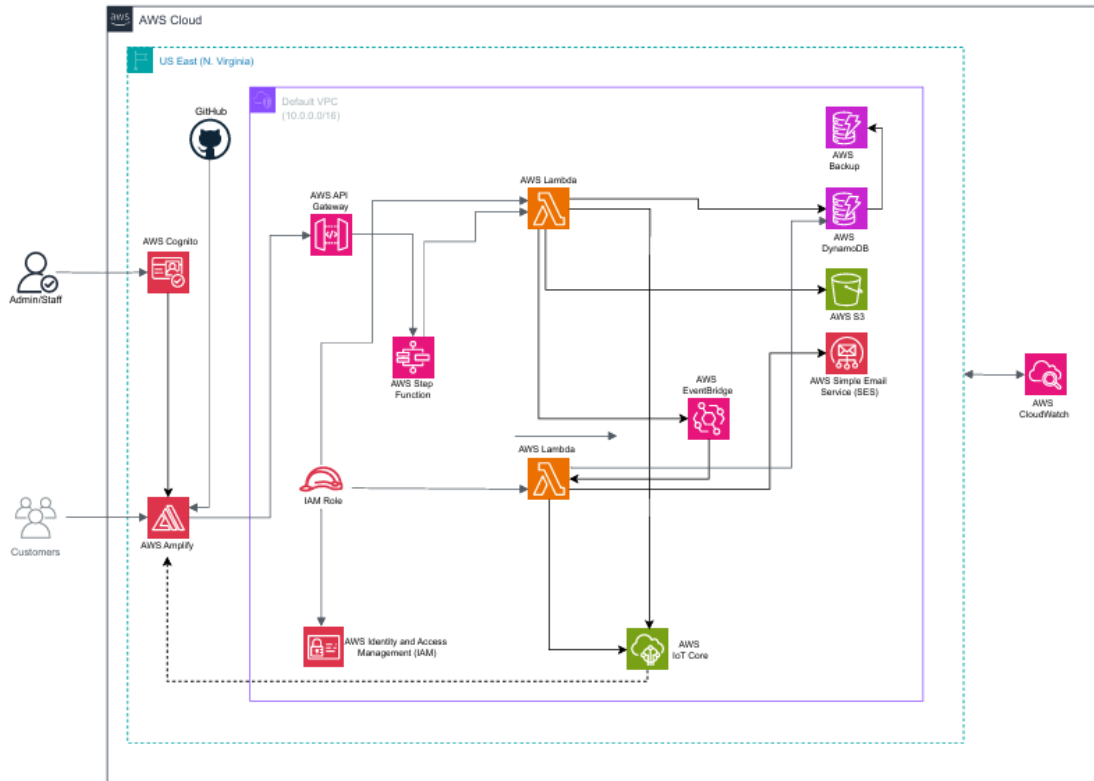


PetStay Architecture Diagram



Description

PetStay is a serverless, event-driven order processing system architected on AWS to efficiently manage pet appointment bookings. The application frontend is hosted on AWS Amplify, enabling pet owners to submit booking requests via a responsive web interface. Amazon Cognito secures authentication for both customers and administrators, enforcing strong password policies and integrating with IAM roles based on the principle of least privilege.

Upon a booking submission, requests are routed through Amazon API Gateway, which acts as a fully managed API front door. These requests initiate an AWS Step Functions state machine that orchestrates the backend workflow. This workflow invokes multiple AWS Lambda functions responsible for:

1. Validating booking details
2. Persisting data in Amazon DynamoDB
3. Generating QR codes and storing them securely in Amazon S3 with AES-256 server-side encryption and public access fully blocked
4. Sending booking confirmation emails via Amazon SES (with email delivery scoped to admin addresses under AWS Free Tier)
5. Emitting structured events to Amazon EventBridge for downstream processing

Administrative actions such as confirming, canceling, restoring, or checking in appointments are managed through a secure admin dashboard. These actions trigger a separate Lambda function that updates booking records and emits additional events to EventBridge, which then propagates real-time notifications using AWS IoT Core for connected systems or user interfaces.

The application infrastructure is deployed within the default AWS Virtual Private Cloud (VPC), spanning six public subnets across multiple Availability Zones (us-east-1a to us-east-1f) to ensure high availability and fault tolerance.

Additional operational and security best practices include:

1. DynamoDB point-in-time recovery (PITR) for data protection and fast restore capabilities
2. Amazon CloudWatch integration for comprehensive logging, monitoring, and operational insights
3. API Gateway access control through scoped usage plans and API keys, limiting access to authorized administrators.